CS 411W Professional Workforce Development II

Lab 2

Requirements & Specifications

Individual Portion

FAICT – Fact AI Checking Tool

Johnovon Richards

Group Blue

Old Dominion University

Professor: Sarah Hosni

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1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) is to define the functional characteristics, system environment, user characteristics, constraints, assumptions, and dependencies of the FAICT system.

This document is intended primarily for developers responsible for designing, implementing, testing, and maintaining FAICT. It provides a common technical reference for the expected behavior of the prototype and describes the major software components required to support that behavior.

The SRS also establishes a foundation for later detailed product specifications and software requirements. Its purpose is to reduce ambiguity during development by documenting the scope of the system, the responsibilities of its components, and the relationships between those components.

1.2 Scope

FAICT, the Fact AI Checking Tool, is a web-based verification system designed to help college students evaluate questionable or misleading claims using externally retrieved evidence and AI-assisted analysis.

The current prototype accepts text-based claims or questions through a web interface. The system retrieves evidence from external sources, prioritizes trusted information, analyzes the submitted information against the retrieved evidence, and presents a structured verification result to the user.

The primary goals of FAICT are to:

- reduce the time required to manually verify questionable online information;
- provide users with visible supporting evidence rather than only an AI-generated answer;
- identify whether available evidence supports, contradicts, qualifies, or cannot establish a claim;
- recognize when a claim depends on additional context before a reliable conclusion can be reached;
- prioritize relevant and authoritative sources;
- provide explanations describing what the evidence establishes and what remains uncertain;
- maintain a history of completed verification requests.

The primary target users of the prototype are undergraduate college students. Example uses include verifying financial-aid information, academic policies, research claims, misleading social-media content, and other information that may influence student decisions.

The current prototype focuses on text-based verification through a web application. Browser-extension support, speech-to-text verification, video analysis, AI-generated-content detection, and advanced user-account functionality are not required for the current prototype.

1.3 Definitions, Acronyms, and Abbreviations

AI Artificial Intelligence.

API Application Programming Interface.

FAICT Fact AI Checking Tool.

GUI Graphical User Interface.

IDE Integrated Development Environment.

LLM Large Language Model.

AI-Generated Content
Text, images, audio, or video created or significantly modified using artificial intelligence systems.

Browser Extension A software component installed within a web browser to provide additional functionality.

Credibility Assessment
A structured evaluation describing how available evidence relates to a submitted claim.

Disinformation False or misleading information intentionally created or distributed to deceive others.

Evidence Retrieval The process of locating external information that may support, contradict, or provide context for a submitted claim.

Fact-Checking The process of evaluating a factual claim by comparing it against reliable external evidence.

Misinformation False or misleading information that may be distributed regardless of intent.

Primary Source An original or authoritative source directly responsible for the information being referenced, such as a government agency, university policy page, or original research publication.

Trusted Source A source considered reliable based on authority, reputation, domain, subject matter, or other verification criteria.

Verification The process of determining whether available evidence supports, contradicts, qualifies, or fails to establish a submitted claim.

Conditional Claim A claim whose correct interpretation depends on additional circumstances, policy conditions, user-specific information, or other contextual factors.

Claim Decomposition The process of separating a complex statement containing multiple factual assertions into smaller claims that may be evaluated independently.

Evidence Trail A stored record containing a submitted claim, sources, source information, analysis result, explanation, and related verification information.

1.4 References

The following sources provide prior project documentation, background information, or technical information relevant to FAICT.

References

[1] J. Richards, "FAICT – Fact AI Checking Tool: Lab 1 Project Outline," CS 411W Professional Workforce Development II, Old Dominion University, Version 1.3, Oct. 2026.

[2] Google, "Fact Check Tools," Google. [Online]. Available: https://toolbox.google.com/factcheck/about

[3] American Psychological Association, "How and why does misinformation spread?" [Online]. Available: https://www.apa.org/topics/journalism-facts/how-why-misinformation-spreads

[4] Snopes, "About Snopes." [Online]. Available: https://www.snopes.com/about/

[5] C. Domonoske, "Fact-checking website Snopes is fighting to stay alive," NPR, July 26, 2017.

[6] B. Cruz and G. Petrino, "90% of people claim they fact-check news stories as trust in media plummets," Security.org, Oct. 22, 2025.

[7] Statista, "Internet usage worldwide – statistics & facts." [Online]. Available: https://www.statista.com/topics/1145/internet-usage-worldwide/

1.5 Overview

The remainder of this SRS describes the overall structure and expected behavior of FAICT.

Section 2 provides an overall description of the system. Section 2.1 describes the product perspective and relationships between the frontend, backend, database, evidence-retrieval services, and AI analysis service.

Section 2.2 summarizes the primary functions provided by the software. Section 2.3 describes the intended users of FAICT. Section 2.4 identifies technical and operational constraints. Section 2.5 describes system assumptions and external dependencies.

Later sections of the SRS will define detailed functional and nonfunctional software requirements.

2. Overall Description

2.1 Product Perspective

FAICT is implemented as a Flask-based monolithic web application written primarily in Python. The prototype combines the frontend interface, backend application logic, database access, external evidence retrieval, and AI-assisted analysis within a single web application.

Although the application follows a monolithic architecture, responsibilities are separated into routes, models, services, templates, and tests.

The frontend is implemented using HTML, Jinja templates, Tailwind CSS, and limited JavaScript. Flask provides backend routing and coordinates communication between the user interface, database, evidence-retrieval service, and AI analysis service.

Flask-SQLAlchemy provides object-relational mapping between Python objects and the SQLite development database. The database currently stores submitted claims and completed analysis results.

The evidence-retrieval subsystem communicates with external services. Google Fact Check Tools is used when relevant formal fact-check information is available. Brave Search is used as a fallback when an applicable formal fact check cannot be found.

Retrieved evidence is normalized into a common internal representation before being passed to the AI analysis subsystem. General web-search results are filtered to favor trusted domains and authoritative sources.

The AI analysis subsystem communicates with the OpenAI API. The AI is used to analyze a submitted claim against evidence retrieved by the system. It is not treated as the original source of factual evidence.

The current system architecture is shown in Figure 1.

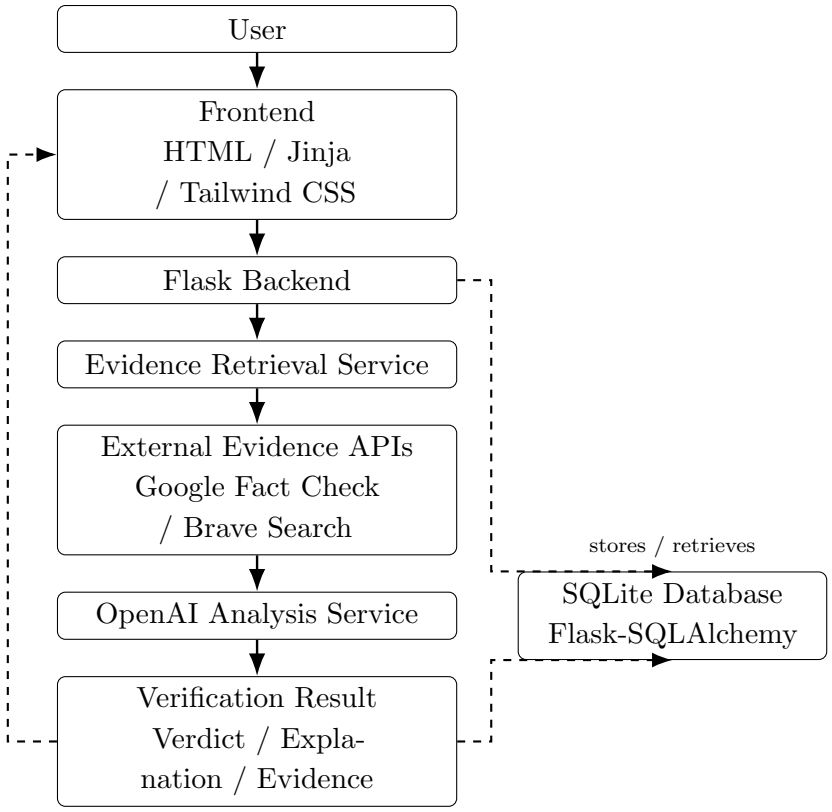


Figure 1: FAICT Major Functional Component Diagram

The architecture separates evidence retrieval from AI analysis. This separation allows evidence providers to be replaced or expanded without requiring significant changes to the frontend or analysis components.

It also reduces reliance on the OpenAI model as a factual source because the analysis is based on externally retrieved evidence.

2.2 Product Functions

The FAICT prototype provides the following primary software functions.

2.2.1 Claim Submission

The system accepts text-based claims or questions through the web interface.

Submitted text is transferred to the Flask backend for processing.

2.2.2 Claim Classification

The system is intended to distinguish between different types of submitted information so that claims may be processed appropriately.

Planned classifications include:

- factual;
- conditional;
- disputed;
- unverifiable.

This functionality is intended to prevent context-dependent claims from being forced into misleading true-or-false results.

2.2.3 Claim Decomposition

Complex statements may contain multiple independent factual assertions.

FAICT is intended to separate these statements into smaller claims when individual verification is required.

For example:

> "If I drop a class, I will lose my scholarship, owe back my FAFSA money, and lose financial aid next semester."

contains multiple assertions that may require separate evidence and conditions.

2.2.4 Evidence Retrieval

FAICT retrieves external evidence related to submitted claims.

The current retrieval process uses:

1. Google Fact Check Tools when applicable;
2. Brave Search when a suitable formal fact-check result is unavailable.

Evidence returned from different services is normalized into a common internal format before additional processing.

2.2.5 Trusted and Authoritative Source Filtering

General web-search results are filtered before being accepted as evidence.

The current system favors recognized:

- government organizations;
- educational institutions;
- scientific organizations;
- medical and health organizations;
- established fact-checking organizations;
- major news organizations;
- other approved authoritative domains.

Domains ending in `.gov`, `.edu`, and `.mil` may also be prioritized when appropriate.

Future versions may apply topic-specific source ranking.

For example, a financial-aid claim may prioritize StudentAid.gov and official university financial-aid resources over general secondary sources.

2.2.6 Evidence Relevance

Source trustworthiness and evidence relevance are treated as separate concepts.

A source may be generally trustworthy without directly addressing the submitted claim.

The system is therefore being developed to evaluate how closely retrieved evidence relates to the submitted information before the evidence is used for analysis.

2.2.7 AI-Assisted Evidence Analysis

The submitted claim and normalized evidence are provided to the OpenAI API for analysis.

The AI service is instructed to base its response on the supplied evidence rather than inventing unsupported sources or citations.

The resulting analysis may include:

- verdict;
- analysis confidence;
- summary;
- explanation;
- relevant conditions or exceptions;
- identification of insufficient evidence.

The currently implemented internal verdict values are:

- Supported;
- Misleading;
- Unsupported;
- Uncertain.

The user interface may display the internal *Unsupported* classification as *Contradicted* when that terminology more clearly communicates the result.

2.2.8 Missing Context and Conditional Results

Some submitted questions cannot be reliably answered without additional information.

The system is intended to identify relevant missing context instead of presenting an unsupported definitive conclusion.

For example, the effect of withdrawing from a university course on financial aid may depend on:

- remaining credit hours;
- enrollment status;
- type of financial aid;

- date of withdrawal;
- institutional or scholarship requirements.

FAICT should communicate these conditions to the user when they affect the reliability of the result.

2.2.9 Uncertainty Handling

If usable evidence cannot be retrieved, the system does not generate an unsupported factual conclusion.

The current prototype may return an *Uncertain* result when sufficient evidence is unavailable.

This distinguishes a lack of evidence from evidence that directly contradicts a claim.

2.2.10 Results Display

The web interface presents the completed verification result to the user.

The results interface may display:

- submitted claim;
- claim type or classification;
- verdict;
- analysis confidence;
- summary;
- explanation;
- supporting evidence;
- source links;
- missing context;
- recommended next steps.

2.2.11 Submission Storage

Completed analyses are stored through Flask-SQLAlchemy using the SQLite development database.

The current Submission model stores:

- submission identifier;
- submitted claim;
- verdict;
- confidence;
- summary;
- explanation;
- creation timestamp.

Future database revisions may store individual evidence records, source metadata, claim classifications, and other verification details.

2.2.12 History

The system provides a History page containing previously saved analyses.

Saved submissions are retrieved from the database and displayed in reverse chronological order.

Future versions may allow a user to reopen a complete verification record containing the original evidence and source details.

2.2.13 Prototype Feature Status

The current prototype implements:

- text-based claim submission;
- external evidence retrieval;
- Google Fact Check integration;
- Brave Search fallback;
- trusted-source filtering;
- evidence normalization;
- AI-assisted evidence analysis;
- verdict, confidence, summary, and explanation output;
- SQLite submission persistence;
- submission History;
- automated backend testing.

Features planned or currently under development include:

- claim classification;
- conditional-claim detection;
- claim decomposition;
- topic-aware source prioritization;
- evidence-relevance scoring;
- source-conflict detection;
- source publication and update dates;
- persistent evidence trails;
- missing-context explanations.

Potential future product features include:

- website and article URL submission;
- browser-extension integration;

- speech-to-text claim submission;
- video transcript analysis;
- AI-generated-content detection;
- user accounts;
- advanced research-verification tools.

2.3 User Characteristics

The primary users of the FAICT prototype are undergraduate college students.

Users are expected to have general web-browsing experience but are not expected to possess technical knowledge of artificial intelligence, software APIs, programming, or professional fact-checking techniques.

Typical student use cases include:

- verifying financial-aid information;
- checking academic or institutional policies;
- evaluating claims encountered through social media;
- checking information before using it in coursework;
- evaluating research-related claims;
- comparing evidence from multiple sources.

FAICT should be usable by students with limited experience evaluating source authority.

Verification results should therefore use clear language and identify the external sources used in the analysis.

Users may submit either direct factual statements or natural-language questions.

The user is not required to understand how the evidence retrieval, database, or AI components operate internally.

FAICT is intended to assist verification and decision-making, but it does not replace official professional, academic, financial, medical, or legal guidance.

For example, when financial-aid eligibility depends on a student's individual circumstances, FAICT may identify relevant policies and conditions but should recommend confirmation with the appropriate university office.

Secondary future users may include:

- researchers;
- educators;
- journalists;
- content creators;
- online moderators;
- general internet users.

2.4 Constraints

FAICT is subject to the following technical and operational constraints.

2.4.1 External API Availability

FAICT depends on external APIs for evidence retrieval and AI analysis.

If Google Fact Check, Brave Search, or OpenAI becomes unavailable, experiences an outage, changes its API, or imposes new restrictions, related functionality may be degraded.

2.4.2 API Usage Limits

Third-party services may enforce request limits, usage quotas, or billing restrictions.

The software must therefore avoid unnecessary requests and handle external API errors without terminating the application.

2.4.3 Internet Connectivity

An active internet connection is required to retrieve current external evidence and communicate with third-party APIs.

2.4.4 Evidence Availability

FAICT cannot guarantee that sufficient evidence exists for every submitted claim.

Some claims may be:

- too personal;
- too recent;
- too vague;
- poorly documented;
- outside the coverage of available evidence providers.

In these situations, the software may return an uncertain or unverifiable result.

2.4.5 Evidence Relevance

A trusted source may still be unrelated to the submitted claim.

The existing trusted-domain filtering mechanism therefore cannot guarantee that every accepted source provides useful evidence.

Improved relevance evaluation is an ongoing prototype requirement.

2.4.6 Prototype Input Limitations

The current prototype primarily supports text-based claims and questions.

Audio, video, image, and browser-extension input are outside the required prototype scope.

2.4.7 Database Scale

SQLite is currently used as the development database.

SQLite provides sufficient functionality for local prototype development but may not satisfy future requirements for large-scale concurrent access.

PostgreSQL may replace SQLite if larger deployment requirements are introduced.

2.4.8 AI Limitations

AI-generated analysis may contain errors or unexpected output.

The application therefore validates AI responses and attempts to restrict analysis to the evidence retrieved by the system.

An AI-generated result should not be treated as authoritative independently of the displayed evidence.

2.4.9 User Information

The prototype should avoid collecting or storing unnecessary personal information.

Users should not be required to provide sensitive financial, academic, medical, or identifying information simply to verify a general claim.

2.5 Assumptions and Dependencies

FAICT operates under the following assumptions and external dependencies:

- The user has access to a modern web browser.
- The system has an active internet connection.
- A compatible Python runtime is installed on the development system.
- Required Python packages are installed from the project's dependency file.
- Flask remains available as the backend web framework.
- Jinja remains available for server-side HTML templating.
- Flask-SQLAlchemy remains available for database access.
- SQLite remains available as the prototype development database.
- Required environment variables and API keys are correctly configured.
- The OpenAI API remains available for AI-assisted analysis.
- Google Fact Check Tools remains available for fact-check retrieval.
- Brave Search remains available as a general web-search fallback.
- External websites referenced as evidence remain publicly accessible.
- Third-party API response formats remain compatible with the implemented integrations, or the software is updated when those formats change.
- Trusted and authoritative sources provide sufficiently accurate information for the claims being evaluated.

- Users understand that FAICT provides evidence-assisted verification rather than guaranteed professional advice.